



FABRIC MANUFACTURING

Part: B

Course Code: FE 253-0723

Quantitative Aptitude

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MATHEMATICAL FORMULA

1. No. of Needles = No. of Wales = $\pi \times \text{Machine Diameter (D)} \times \text{Machine Gauge (E)}$
2. No. Feeders = No. of cam-box = No. of Courses/rev.
3. Courses Per Day (CPD) = RPM (n) x Feeder (F) x Efficiency (η) x Time (60 x 24)
4. Fabric Length (m/Day) =
$$\frac{\text{Course Per Day}}{\text{Feeders Per Course} \times \text{Courses Per Centimeter (CPC)} \times 100} = \frac{nF\eta t}{FPC \times CPC \times 100}$$
5. Fabric Width (m) =
$$\frac{\text{No of Wales}}{\text{Wales Per Centimeter (WPC)} \times 100} = \frac{\pi DE}{WPC \times 100}$$
6. GSM =
$$\frac{WPC \times CPC \times 100 \times 100 \times SL \times Tex}{1000 \times 1000}$$
7. Production (Kg/Day) =
$$\frac{CPD \times CL \times Tex}{1000 \times 1000 \times 1000}$$
8. Course Length (CL) = Stitch Length (SL) x π DE

PROBLEM - 1

Calculate the length in meters of a plain, single sided or single jersey fabric knitted at 20 courses/cm on 30” diameter 22-gauge circular machine having 108 feeds. The machine operates for 8 hours at 36 rpm at 87% efficiency.

Solution:

Given,

RPM, $n = 36$

Gauge, $E = 22$

Diameter, $D = 30$ inch

Feeder, $F = 108$

Efficiency, $\eta = 87\%$

Course per cm, $CPC = 20$

Feeders per Course = 1

$$\begin{aligned}\text{Fabric Length, FL} &= \frac{\text{Course Per Day}}{\text{Feeders Per Course} \times \text{Courses Per Centimeter (CPC)} \times 100} \\ &= \frac{nF\eta t}{FPC \times CPC \times 100} \\ &= \frac{36 \times 108 \times 0.87 \times 8 \times 60}{1 \times 20 \times 100} \\ &= 811.82 \text{ m. (Ans).}\end{aligned}$$

PROBLEM - 2

Calculate the length in meters of a plain, single sided or single jersey fabric knitted at 16 courses/cm on 26” diameter 28-gauge circular machine having 104 feeds. The machine operates for 8 hours at 29 rpm at 95% efficiency.

Solution:

Given,

RPM, $n = 29$

Feeder, $F = 104$

Efficiency, $\eta = 95\%$

Gauge, $E = 28$

Diameter, $D = 26$ inch

Course per cm, $CPC = 16$

Feeders per Course = 1

$$\begin{aligned}\text{Fabric Length, FL} &= \frac{\text{Course Per Shift}}{\text{Feeders Per Course} \times \text{Courses Per Centimeter (CPC)} \times 100} \\ &= \frac{nF\eta t}{FPC \times CPC \times 100} \\ &= \frac{29 \times 104 \times 0.95 \times 8 \times 60}{1 \times 16 \times 100} \\ &= 859.56 \text{ m. (Ans).}\end{aligned}$$

PROBLEM - 3

If the cylinder diameter is 30”, machine gauge is 32 and the wales per cm is 14. Fabric width?

Solution:

Given,

Gauge, E = 32

Diameter, D = 30 inch

Wales per cm, WPC = 14

$$\text{Fabric Width, FW} = \frac{\text{No of Wales}}{\text{Wales per cm (WPC)} \times 100}$$

$$= \frac{\pi DE}{WPC \times 100}$$

$$= \frac{\pi \times 30 \times 32}{14 \times 100}$$

$$= 2.153 \text{ m. (Ans).}$$

PROBLEM - 4

A single jersey circular knitting machine of 28G & 30" has 90 Feeders. It produces plain jersey structure using 34 Ne yarn with a stitch length of 2.7 mm. If the cylinder of the machine is revolved at 28 rpm, calculate the production in Kg per shift considering 78% efficiency.

Solution:

Given,

RPM, $n = 28$

Stitch Length, $SL = 2.7$ mm

Gauge, $E = 28$

Diameter, $D = 30$ inch

Feeder, $F = 90$

Course Length, $CL = SL \times \text{No of Needles}$

$$= SL \times \pi DE$$

$$= 2.7 \times 3.1412 \times 30 \times 28$$

$$= 7124.24 \text{ mm}$$

Course Per Shift, $CPS = nF\eta t$

$$= 28 \times 90 \times 0.78 \times 8 \times 60$$

$$= 943488$$

PROBLEM - 4

A single jersey circular knitting machine of 28G & 30" has 90 Feeders. It produces plain jersey structure using 34 Ne yarn with a stitch length of 2.7 mm. If the cylinder of the machine is revolved at 28 rpm, calculate the production in Kg per shift considering 78% efficiency.

$$\text{Tex} = 590.5/34 = 17.37$$

$$\begin{aligned}\text{Production (Kg/Shift)} &= \frac{CPS \times CL \times Tex}{1000 \times 1000 \times 1000} \\ &= \frac{943488 \times 7124.24 \times 17.37}{1000 \times 1000 \times 1000} \\ &= 116.74 \text{ kg/Shift}\end{aligned}$$

Ans: 116.74 Kg/Shift.

PROBLEM - 5

A single jersey fabric has WPI = 34, CPI = 42 and a stitch length of 2.83 mm. If the fabric is knitted with 28 Ne cotton yarn, calculate the GSM of the fabric.

Solution:

Given,

$$\text{WPI} = 34 \text{ So, } \text{WPC} = 34/2.54 = 13.385$$

$$\text{CPI} = 42 \text{ So, } \text{CPC} = 42/2.54 = 16.535$$

$$\text{SL} = 2.83 \text{ mm}$$

$$\text{Ne} = 28 \text{ So, } \text{Tex} = 590.5/28 = 21.089$$

$$\begin{aligned} \text{GSM} &= \frac{\text{WPC} \times \text{CPC} \times 100 \times 100 \times \text{SL} \times \text{Tex}}{1000 \times 1000} \\ &= \frac{13.385 \times 16.535 \times 100 \times 100 \times 2.83 \times 21.089}{1000 \times 1000} \\ &= 132.08 \end{aligned}$$

Ans: 132.08

PROBLEM - 6

Values of Circular knitting machine:

RPM, $n = 31$

Gauge, $E = 28$

Diameter, $D = 30$ inch

Feeder, $F = 96$

Efficiency, $\eta = 85\%$

Values of article:

Structure: Plain Interlock

Yarn: PET 76 dTex

CPC: 17 courses/cm

WPC: 14 wales/cm

Fabric Weight, GSM = 100 gm/sq. m

$$\begin{aligned}\text{Fabric Length, FL} &= \frac{\text{Course Per Hr}}{\text{Feeders Per Course} \times \text{Courses Per Centimeter (CPC)} \times 100} \\ &= \frac{nF\eta t}{FPC \times CPC \times 100} = \frac{31 \times 96 \times 0.85 \times 60}{2 \times 17 \times 100} = 44.64 \text{ m/Hr.}\end{aligned}$$

PROBLEM - 6

$$\begin{aligned}\text{Fabric Width, FW} &= \frac{\text{No of Wales}}{\text{Wales per cm (WPC)} \times 100} \\ &= \frac{\pi DE}{WPC \times 100} \\ &= \frac{\pi \times 30 \times 28}{14 \times 100} \\ &= 1.88 \text{ m. (Ans).}\end{aligned}$$

$$\begin{aligned}\text{Production,} &= \frac{FL \left(\frac{m}{Hr} \right) \times FW (m) \times GSM \left(\frac{gm}{m^2} \right)}{1000} \\ &= \frac{44.64 \times 1.88 \times 100}{1000} \\ &= 8.39 \text{ Kg/Hr. (Ans).}\end{aligned}$$

THANK
YOU !

